Research Paper Implementation Plan

Graph Digital Twin Project Guide

August 4, 2026

Contents

Decoupling Cyber-Physical Grids and Collaborative Industrial Processes: A Refined Implementation Roadmap for Adaptive Explainable Digital Twins

The integration of advanced communication networks and computational intelligence into physical utility systems has established highly complex Cyber-Physical Power Systems.1 While these digital architectures improve operational efficiency, they expand the attack surface of critical infrastructure, exposing them to sophisticated, multi-vector cyber-physical threats.2
An evaluation of the initial research proposal reveals a common limitation: the conflation of the smart grid environment with the Industry 5.0 collaborative manufacturing paradigm.3 Although both concepts share enabling technologies such as the Internet of Things, edge computing, and digital twins, they operate under fundamentally distinct physics, constraints, and regulatory standards.3
To build a scientifically rigorous and defensible research framework suitable for high-impact IEEE Transactions, these domains must be decoupled.3 This technical report provides a comprehensive, comparative analysis of two refined research paths—the System-Level Smart Grid Twin and the Process-Level Continuous Manufacturing Twin—answering the architectural and physical constraints of each track to prepare the researcher for the upcoming advisory defense.3

Conceptual Separation of Smart Grids and Industry 5.0

Smart grids are highly synchronous, continuous physical networks governed by the laws of electromagnetism, transient dynamics, and strict grid-synchronization codes.8 Industrial control within these systems requires sub-millisecond to millisecond-level telemetry and actuation to prevent catastrophic physical instability, such as voltage collapses or out-of-step generator trippings.8
Conversely, Industry 5.0 represents a manufacturing paradigm focused on collaborative human-machine workspaces, custom manufacturing lines, and process optimization within localized, discrete, or continuous industrial plants.6 Combining these domains into a single research question introduces analytical ambiguity, as the system boundaries, threat models, and real-time execution constraints are fundamentally irreconcilable.3

Feature	Substation-Level Smart Grid (Path A)	Continuous Proc
Primary Paradigm	High-fidelity Cyber-Physical Power Systems 1	Industry 5.0 Hu
Governing Physical Laws	Kirchhoff's Laws, AC power flow equations 12	Mass balances, c
Primary System Boundary	Substation, distribution feeder, or microgrid 16	Automated chem
Typical Sampling Rates	10 to 60 samples per second (PMU / Synchrophasors) 9	1 sample per 1 t
Critical Actuation Latency	Sub-cycle to millisecond range 9	Second to minut
Core Vulnerability Target	AC state estimation and bad data detection (BDD) 12	Multivariate sens
Target IEEE Journal	*IEEE Transactions on Smart Grid* 8	*IEEE Transactio*

To address the advisory feedback, two alternative implementations have been designed.3 Path A focuses on a system-level smart grid digital twin modeling cyber-physical interdependencies 1, while Path B explores a process-level digital twin of a continuous chemical manufacturing plant using the standard Tennessee Eastman Process.3

Path A Implementation: Substation-Level Smart Grid System Digital Twin

Path A refines the student's original focus, centering on a localized, cyber-physical smart substation or microgrid system.3 It models the system-level interdependencies between communication

networks and physical distribution feeders.2

Mathematical State Estimation and Threat Model

The physical power system is modeled as an undirected, weighted graph:
![][image1]
where the set of vertices ![][image2] corresponds to electrical buses, the set of edges ![][image3] represents transmission branches, and ![][image4] is derived from the real part of the nodal admittance matrix ![][image5].23 The physical states of the system—comprising voltage magnitudes ![][image6] and voltage angles ![][image7] at all buses—are estimated from noisy SCADA measurements.12 This relationship is formulated as:
![][image8]
where ![][image9] is the measurement vector (including active and reactive power injections and flows), ![][image10] is the system state vector, ![][image11] represents the non-linear power flow equations, and ![][image12] is the Gaussian measurement noise.12
An adversary targeting the substation state estimator designs a False Data Injection Attack (FDIA) vector ![][image13].12 To bypass traditional Bad Data Detection (BDD) algorithms that analyze the largest normalized residual, the attacker construct ![][image14] such that 13:
![][image15]
where ![][image16] is a vector of malicious state deviations.12 Consequently, the corrupted measurement vector ![][image17] yields a residual that remains below the detection threshold, rendering the attack unobservable to classical estimators.12

Digital Twin Architecture and Components

To detect and mitigate these unobservable attacks, a hybrid, hardware-in-the-loop (HIL) digital twin is required.5 The virtual twin is continually synchronized with physical assets using a robust telemetry pipeline.29

———————————————————————————+

PHYSICAL LAB-BASED SETUP

TYPHOON HIL / OP5707

IED / SCADA EMULATOR

(IEEE 14-Bus System Solver)

(Raspberry Pi 4B)

Executes AC transients at

Reads electrical signals via

a sub-50 microsecond step

—

v

———————————————————————————+

VIRTUAL DIGITAL TWIN LAYER

GNN CLASSIFIER

EXPLAINABILITY ENGINE

(GraphSAGE / PyG)

(GNNExplainer & SHAP)

Processes spatial dependencies

Maps structural anomalies and

to flag anomalous states

attributes back to operators

———————————————————————————+

The hardware setup is designed as a Controller Hardware-in-the-Loop (CHIL) platform.31

- **Simulator (Physical Emulation)**: The physical grid dynamics of an IEEE 14-bus test system are executed in real time on a **Typhoon HIL 402** or **OPAL-RT OP5707XG** simulator at a time step of 40 ![][image18].32
- **Local Controllers (Physical Components)**: **Raspberry Pi 4B** units act as physical substation Intelligent Electronic Devices (IEDs).31 These devices interface with the simulator using physical Ethernet cables, executing standard communication protocols such as Modbus TCP and MQTT to poll the simulator.32
- **Software Representation**: The virtual replica models the **entire system** (nodes, branches, and communication channels).1 It executes a Python backend using PyTorch Geometric to deploy a spatiotemporal Graph Neural Network (GraphSAGE or GATv2) that parses the system's admittance topology and detects relational anomalies.38

To establish the physical validation platform, the specific hardware and software components outlined below must be procured.

Component Group	Component Name	Technical Specifications
Real-Time Solver	OPAL-RT OP5707XG	FPGA-based real-time simulation, eHS solver
Physical IED Nodes	Raspberry Pi 4 Model B	Broadcom BCM2711, 4GB LPDDR4, Gigabit Eth
HIL Interface	uGrid DSP Interface	12-bit Analog-to-Digital Channels, 3.3V I/O limits
Communications	Modbus Slave Driver	RT-LAB native async driver, TCP Port 502
Network Switch	Unmanaged Industrial gigabit	8-Port RJ45, DIN-Rail, 24V DC input
Graph Platform	PyTorch Geometric	v2.5.0, CUDA-accelerated graph-batching

The GNN Anomaly Classifier and Explainability Layer

The edge-polling script on the Raspberry Pi aggregates physical measurement snapshots at 1 Hz and structures them as graph node feature matrices ![][image19], containing ![][image20] for each bus.17
The GNN classifier consists of a two-layer GraphSAGE architecture designed to process spatial correlations within the non-Euclidean power grid network, outperforming flat classifiers that fail to capture the system's topological constraints.13 The mathematical representation of the aggregation and update steps at layer ![][image21] is:
![][image22]
where ![][image23] is the feature representation of bus ![][image24], ![][image25] represents the physical transmission connections to bus ![][image24], and ![][image26] denotes concatenation.40
To interpret these predictions for substation operators, a dual-layer explainability framework is deployed:

- **Topological Subgraph Explanation (GNNExplainer)**: Upon classifying a state as an anomaly, GNNExplainer computes edge and feature masks by optimizing the mutual information between the GNN's prediction and the compact subgraph structure 38:

![][image27]
where ![][image28] and ![][image29] represent the edge and feature masks, respectively, and ![][image30] denotes element-wise multiplication.38 This identifies the targeted transmission lines and corrupted PMUs, allowing operators to prioritize physical inspections.38

- **Local Feature Attribution (SHAP)**: A parallel TreeSHAP engine attributes Shapley values to individual telemetry registers.45 This quantifies the exact contribution of each

sensor measurement (e.g., active power at bus 9 to the anomaly alarm.45 SHAP is chosen over LIME due to LIME's extreme mathematical instability under slight parameter variations, which can return contradictory explanations for similar inputs and erode operator trust in safety-critical contexts.46

Path B Implementation: Tennessee Eastman Process-Level Industry 5.0 Digital Twin

If the researcher pivots toward a simpler, well-defined Industry 5.0 paradigm as suggested by the advisor, the **Tennessee Eastman Process (TEP)** represents an ideal benchmark.3 This track shifts the research focus from a distributed power system to a centralized, continuous industrial process.14

Process Boundary and Mathematical Formulation

The TEP modeled by Downs and Vogel in 1993 represents a high-fidelity simulation of an actual chemical manufacturing process.14 The process contains **8 chemical components** (4 reactants, 2 products, 1 by-product, and 1 inert component) operating across **5 primary process units**: a chemical reactor, a product condenser, a gas-liquid separator, a product stripper, and a centrifugal compressor.14
The system state is characterized by ![][image31] continuously monitored variables (including temperatures, pressures, and flowrates) and ![][image32] manipulated variables (valve positions, reaction agitator speed), resulting in a 53-dimensional multivariate time-series vector at each sampling step ![][image33] 14:
![][image34]
Anomalies in this process arise from 20 simulated plant faults (including valve stickings, feed temperature variations, and cooling water flow failures) or cyber-physical malicious interventions targeting the manipulated variables to trigger physical shutdowns.14
To align this continuous process with the human-centric focus of Industry 5.0, the digital twin implements a **Virtual Mass Balance (VMB)** framework.15 The VMB utilizes virtual soft sensors to continuously calculate materials balance tables, enriching SCADA logs with real-time thermodynamic constraints 15:
![][image35]
where ![][image36] represents mass flow rates, ![][image37] is the reactor mass hold-up, and ![][image38] is a dynamic residual indicator of unmodeled dynamics or sensor tampering.15

Digital Twin Architecture and Components

Unlike the distributed substation setup of Path A, the TEP digital twin is primarily **data-driven**, modeling a **process** rather than an entire distributed infrastructure.14



(Downs & Vogel Model)

(Docker)

Generates multivariate time-

Emulate industrial SCADA and

series process data

network architectures



ATTENTION CLASS-FORECAST

EXPLAINABILITY ENGINE

(TransKal / LSTM)

(TreeSHAP)

Performs joint state tracking

Isolates the exact root-cause

and anomaly detection

component or valve

+

- **Physical Representation (Software Emulated)**: The chemical process dynamics are emulated using a high-fidelity Python or MATLAB simulation of the TEP.14 To add a realistic communication layer without physical hardware, the control loops are containerized using **Docker** containers running open-source programmable logic controller (OpenPLC) software, interconnected via virtual network adapters using **Containernet**.22
- **Physical Components (Workstation-based)**: This track requires **no physical laboratory hardware** beyond a high-performance workstation equipped with a modern multi-core CPU and a GPU (such as an NVIDIA RTX 4070 or higher) to accelerate deep sequence models.21
- **Scale of Modeling**: The digital twin is designed at the **Process-Level**.14 It mirrors the thermal and fluid interactions of the 5 coupled units, providing an detailed virtual representation of a continuous plant's operation.14

The software stack required for this data-driven process twin is detailed in the table below.

Software Layer	Specific Package / Library	Role in Architecture
TEP Core Solver	TEP Python Simulator	Emulates the 53 multivariate process streams
Control Emulation	Docker / OpenPLC	Containerized PLCs executing ladder logic controls
Virtual Networking	Containernet	Emulates industrial network delays and attacks
Core AI Library	PyTorch / Scikit-Learn	Compiles the sequential classifier and soft sensors
Model Explainability	SHAP (v0.45.0)	TreeSHAP calculation for feature attribution
Process Monitoring	AVEVA PI / Grafana	Visual dashboard for human-centric monitoring

The Sequential Anomaly Detector and Diagnostic Layer

Because TEP data represents continuous multivariate time series with temporal dependencies, the digital twin leverages a **Transformer-Kalman (TransKal)** classifier.21 This model combines a self-attention mechanism to capture long-range temporal dependencies with a Kalman Filter module to smooth sensor noise and stabilize error signals.21
The input sequence of process variables ![][image39] (where ![][image40] is batch size and ![][image41] is the history window size) is projected into a hidden embedding space:
![][image42]
where ![][image43] are learnable projection matrices.17 The temporal representations are then decoded via a fully connected classification layer to predict the probability of plant faults.17
In the Industry 5.0 paradigm, the worker is positioned at the center of the control loop.6 Therefore, the digital twin must not only trigger an anomaly alarm but also diagnose the root cause of the fault.14 Upon flagging an anomalous process state, the twin executes a localized explainability routine:

- **TreeSHAP Diagnostics**: For tree-based ensemble models (such as XGBoost or ExtraTrees trained on the hidden states ![][image44]), TreeSHAP calculates exact, game-theoretically optimal feature attributions in polynomial time.45
- **Operator Action Mapping**: The calculated Shapley values are displayed on a human-centric operator dashboard, mapping the exact process variables (e.g., reactor cooling water flow rate) that contributed to the alarm.14 This transparent mapping provides industrial operators with actionable diagnostic explanations, mitigating the distrust associated with opaque black-box AI systems.14

Strategic Decision Matrix for the Thursday Advisory Meeting

To guide the academic meeting scheduled for Thursday, this section compiles a comparative decision matrix evaluating Path A (Smart Grid System Twin) and Path B (Tennessee Eastman Process Twin).3 This comparison assesses both tracks across technical risk, development complexity, resource requirements, and academic publication potential.3

Decision Criterion	Path A: Substation-Level Smart Grid
Supervisor's Alignment	Addresses "cyber resilience" and "attack detection" in power networks.3
Model Scale	**System-Level**: Models multiple buses, branches, and communication links.
Primary Approach	**Physical/HIL**: Requires real-time hardware simulators and edge devices.31
Technical Risks	Modbus communication delays, packet loss, and hardware sync issues.29
Hardware Required	OPAL-RT simulator, Raspberry Pi 4B units, Network switch.31
Development Time	High (est. 6–8 months for hardware-in-the-loop setup).54
Dataset Availability	Generated via pandapower or SCADA testbed logs.38
Target IEEE Transactions	*IEEE Transactions on Smart Grid* 8

Recommended Technical Stance for the Advisory Meeting

The student should approach the Thursday meeting with a clear, defensible decision based on resource availability and academic timelines.3
If the institutional laboratory has immediate, unallocated access to an **OPAL-RT** or **Typhoon HIL** real-time simulator, **Path A (Substation-Level Smart Grid)** should be pursued.31 This path provides the high-fidelity, hardware-validated results highly valued by reviewers for *IEEE Transactions on Smart Grid*.7 The student can present the structured bill of materials, the AC state estimation formulation, and the CHIL network topology to demonstrate a concrete, mathematically rigorous, and feasible research scope.3
If real-time simulation hardware is unavailable or if the student has a shorter timeline (e.g., under 6 months), they should pivot to **Path B (Tennessee Eastman Process)**.14 The student can justify this selection to their advisor as a direct response to their feedback, moving to a well-defined process-level industrial digital twin.3 By leveraging containerized PLC nodes and the open-source TEP dataset, the student can build a highly interpretable, data-driven security framework that avoids the overhead of physical hardware validation while maintaining the technical rigor required for *IEEE Transactions on Industrial Informatics*.3

Works cited

- Cyber-Physical Power System Digital Twins—A Study on the State of the Art MDPI, accessed on June 30, 2026, https://www.mdpi.com/1996-1073/18/22/5960

- Cyber-Physical Fusion for GNN-Based Attack Detection in Smart Power Grids, accessed on June 30, 2026, [https://www.researchgate.net/publication/394181729
$_{C}yber - Physical$
$_{F}usion$
$_{f}or$
$_{G}NN - Based$
$_{A}ttack$
$_{D}etection$
$_{i}n$
$_{S}mart$
$_{P}ower$
$_{G}rids$](*https* : *//www.researchgate.net/publication/*394181729*_Cyber−Physical_Fusion_for_GNN−Based_Attack_Detection_in_Smart_Power_Grids*)
- IIIT JIYA.pdf
- Exploring the emerging role of large language models in smart grid cybersecurity: a survey of attacks, detection mechanisms, and mitigation strategies Frontiers, accessed on June 30, 2026, [https://www.frontiersin.org/journals/energy-research/articles/10.3389/fenrg.2025.1531655/full](h research/articles/10.3389/fenrg.2025.1531655/full)

Reliable Microgrid Simulator Systems

- UST Explainers What is Industry 5.0?, accessed on June 30, 2026, https://www.ust.com/en/ust-explainers/what-is-industry-five-point-zero
- IEEE Transactions on Power Systems submission guide Manusights, accessed on June 30, 2026, https://manusights.com/blog/ieee-transactions-on-power-systems-submission-guide
- IEEE Transactions On Smart Grid Overview The Electricity Forum, accessed on June 30, 2026, [https://electricityforum.com/td/smart-grid/ieee-transactions-on-smart-grid](https://electricityfor grid/ieee-transactions-on-smart-grid)
- Smart Grid and Its Requirements on Networking GitHub Pages, accessed on June 30, 2026, [https://leboudec.github.io/leboudec/resources/STW2022.pdf](https://leboudec.github.io/leboude
- Industry 5.0 Explained: From Human-Machine Rivalry to Partnership Simio, accessed on June 30, 2026, https://www.simio.com/blog/industry-5-0-explained-from-human-machine-rivalry-to-partnership
- What is the difference b/w industry 4.0 & industry 5.0? ResearchGate, accessed on June 30, 2026, [https://www.researchgate.net/post/What
$_{i}s$
$_{t}he$
$_{d}ifference$
$_{b}$
$_{w}$
$_{i}ndustry$
$_{4}0$
$_{i}ndustry$

$_5$0](*https* : *//www.researchgate.net/post/What_is_the_difference_b_w_industry_*40*_industry_*50)

- A Comprehensive Survey on the Usage of Machine Learning to Detect False Data Injection Attacks in Smart Grids IEEE Computer Society, accessed on June 30, 2026, [https://www.computer.org/csdl/journal/oj/2025/01/11063250/281HuWfoELe](https://www.computer.
- Graph Neural Networks Based Detection of Stealth False Data Injection Attacks in Smart Grids SciSpace, accessed on June 30, 2026, https://scispace.com/pdf/graph-neural-networks-based-detection-of-stealth-false-data-vei7eyfp.pdf
- The Tennessee Eastman Process: an Open Source Benchmark KeepFloyding, accessed on June 30, 2026, [https://keepfloyding.github.io/posts/Ten-East-Proc-Intro/](https://keepfloyding.github.i East-Proc-Intro/)
- Development of Digital Twins for Continuous Processes: Concept Description of Virtual Mass Balance Based on the Tennessee Eastman Process MDPI, accessed on June 30, 2026, https://www.mdpi.com/2227-9717/13/2/337
- A Graph-Net with Node Embeddings to Detect False Data Injection Attacks in Photovoltaic Systems OSTI, accessed on June 30, 2026, [https://www.osti.gov/servlets/purl/3002204](https://
- A Hierarchical Spatio-Temporal Graph Attention Network for False Data Injection Attack Detection in Smart Grids MDPI, accessed on June 30, 2026, https://www.mdpi.com/2227-9717/14/3/507
- Digital Twin-Based Cyber-Attack Detection and Mitigation for DC Microgrids ResearchGate, accessed on June 30, 2026, [https://www.researchgate.net/publication/385451332 $_D igital$ $_T win - Based$ $_C yber - Attack$ $_D etection$ $_a nd$ $_M itigation$ $_f or$ $_D C$ $_M icrogrids$](*https* : *//www.researchgate.net/publication/*385451332*_Digital_Twin−Based_Cyber− Attack_Detection_and_Mitigation_for_DC_Microgrids*)
- Dataset of Manipulations on the Tennessee Eastman Process Carnegie Mellon University, accessed on June 30, 2026, [https://kilthub.cmu.edu/articles/dataset/Dataset $_o f$ $_M anipulations$ $_o n$ $_t he$ $_T ennessee$ $_E astman$ $_P rocess$/23805552](*https* : *//kilthub.cmu.edu/articles/dataset/Dataset_of_Manipulations_on_the_Ten*
- FAQs for Transactions Authors IEEE Power & Energy Society, accessed on June 30, 2026, [https://ieee-pes.org/publications/transactions-authors-faqs/](https://ieee-pes.org/publications/transact authors-faqs/)

- Benchmarking Machine Learning Fault Detection Methods on the Tennessee Eastman Process Dataset — ChemRxiv, accessed on June 30, 2026, [https://chemrxiv.org/doi/10.26434/chemrxiv.1000
- a cyber-physical digital twin approach to replicating realistic multi-stage cyberattacks on smart grids arXiv, accessed on June 30, 2026, [https://arxiv.org/pdf/2412.04900](https://arxiv.org/pdf/2
- Cyberattack Detection in Large-Scale Smart Grids using Chebyshev Graph Convolutional Networks arXiv, accessed on June 30, 2026, [https://arxiv.org/pdf/2112.13166](https://arxiv.org/pdf/21
- Joint Detection and Localization of Stealth False Data Injection Attacks in Smart Grids using Graph Neural Networks ResearchGate, accessed on June 30, 2026, [https://www.researchgate.net/pub
$_{J}oint$
$_{D}etection$
$_{a}nd$
$_{L}ocalization$
$_{o}f$
$_{S}tealth$
$_{F}alse$
$_{D}ata$
$_{I}njection$
$_{A}ttacks$
$_{i}n$
$_{S}mart$
$_{G}rids$
$_{u}sing$
$_{G}raph$
$_{N}eural$
$_{N}etworks$](*https* : *//www.researchgate.net/publication*/351105381*_Joint_Detection_and_Localization_*
- A GRAPH NEURAL NETWORK MULTI-TASK LEARNING-BASED APPROACH FOR DETECTION AND LOCALIZATION OF CYBERATTACKS IN SMART GRIDS Abdulrah, accessed on June 30, 2026, [https://web1.eng.famu.fsu.edu/ textasciitildetakiddin/publications/paper20.pdf](https://web1.eng.famu.fsu.edu/~takiddin/publications/
- Joint Detection and Localization of Stealth False Data Injection Attacks in Smart Grids using Graph Neural Networks Thomas Overbye, accessed on June 30, 2026, [https://overbye.engr.tamu.edu/ content/uploads/sites/146/2021/10/TSG3117977
$_{p}re$
$_{A}RCHIVE.pdf$](*https* : *//overbye.engr.tamu.edu/wp−content/uploads/sites*/146/2021/10/*TSG*31179
- Generalized Graph Neural Network-Based Detection of False Data Injection Attacks in Smart Grids Katherine Davis, accessed on June 30, 2026, [https://katedavis.engr.tamu.edu/wp-content/uploads/sites/180/2023/02/Generalized
$_{G}raph$
$_{N}eural$
$_{N}etwork - Based$
$_{D}etection$
$_{o}f$
$_{F}alse$
$_{D}ata$
$_{I}njection$
$_{A}ttacks$

$_{i}n$
$_{S}mart$
$_{G}rids{-}1.pdf](https://katedavis.engr.tamu.edu/wp{-}content/uploads/sites/180/2023/02/Generalized$
$Based_Detection_of_False_Data_Injection_Attacks_in_Smart_Grids - 1.pdf)$

- Attention-Enhanced Graph Filtering for False Data Injection Attack Detection and Localization arXiv, accessed on June 30, 2026, [https://arxiv.org/html/2601.18981v1](https://arxiv.org/html/
- Digital Twin vs Simulation — OPAL-RT, accessed on June 30, 2026, https://www.opal-rt.com/blog/digital-twin-vs-simulation/

Real-Time Digital Twinning Solutions	Scalable Digital Twin Technology Tools

- Real-Time Co-Simulation for DC Microgrid Energy Management with Communication Delays arXiv, accessed on June 30, 2026, [https://arxiv.org/html/2511.07052v1](https://arxiv.org/html/
- Real-Time Energy Management in Microgrids: Integrating T-Cell Optimization, Droop Control, and HIL Validation with OPAL-RT MDPI, accessed on June 30, 2026, [https://www.mdpi.com/1 1073/18/15/4035](https://www.mdpi.com/1996-1073/18/15/4035)
- OPAL-RT Technologies Inc System Integration Services MathWorks, accessed on June 30, 2026, [https://www.mathworks.com/products/connections/product
$_{d}etail/opal{-}rt{-}system{-}integration.html](https://www.mathworks.com/products/connections/prod$
$rt - system - integration.html)$
- Smart5Grid Solutions for enhanced TSO grid observability and manageability in massive RES penetration environment PMC, accessed on June 30, 2026, [https://pmc.ncbi.nlm.nih.gov/articles/P
- D9.5 – Hardware-in-the-loop test environment and guidelines for demonstrating non-selective protection systems for meshed HVDC grid PROMOTioN, accessed on June 30, 2026, [https://www.promotion-offshore.net/fileadmin/PDFs/D9.5
$_{H}ardware{-}in{-}the{-}loop_{t}est_{e}nvironment_{a}nd_{g}uidelines_{f}or_{d}emonstrating_{n}on{-}selective_{p}rotection_{s}ystems_{f}or_{m}eshed_{H}VDC_{g}rids.p$
- (PDF) Real-Time Energy Management in Microgrids: Integrating T-Cell Optimization, Droop Control, and HIL Validation with OPAL-RT ResearchGate, accessed on June 30, 2026, [https://www.researchgate.net/publication/394127971
$_{R}eal - Time$
$_{E}nergy$
$_{M}anagement$
$_{i}n$
$_{M}icrogrids$
$_{I}ntegrating$
$_{T} - Cell$
$_{O}ptimization$
$_{D}roop$
$_{C}ontrol$
$_{a}nd$
$_{H}IL$
$_{V}alidation$
$_{w}ith$
$_{O}PAL{-}RT](https://www.researchgate.net/publication/394127971_Real{-}Time_Energy_Managemen$
$Cell_Optimization_Droop_Control_and_HIL_Validation_with_OPAL - RT)$
- Modbus Slave RT-LAB Documentation OPAL-RT, accessed on June 30, 2026, [https://opal-rt.atlassian.net/wiki/spaces/PRD/pages/144215818/Modbus+Slave](https://opal-rt.atlassian.net/wiki/s

- Topology-Aware Cyberattack Detection in Smart Grids Using Explainable Graph Neural Networks — Request PDF ResearchGate, accessed on June 30, 2026, [https://www.researchgate.net/publ
$_{T}opology - Aware$
$_{C}yberattack$
$_{D}etection$
$_{i}n$
$_{S}mart$
$_{G}rids$
$_{U}sing$
$_{E}xplainable$
$_{G}raph$
$_{N}eural$
$_{N}etworks](https://www.researchgate.net/publication/404040073_Topology-Aware_Cyberattack_Det$

- Graph Neural Networks for Electricity Load Forecasting arXiv, accessed on June 30, 2026, https://arxiv.org/html/2507.03690v3

Skills Directory

- RT Box card for studying the control communication impacts on microgrid performance and stability PMC, accessed on June 30, 2026, [https://pmc.ncbi.nlm.nih.gov/articles/PMC9184748/](htt

- DSTF-GKAN: A lightweight spatiotemporal fusion framework for real-time eavesdropping detection in dynamic smart grid networks PMC, accessed on June 30, 2026, [https://pmc.ncbi.nlm.nih.gov

- A Comprehensive Review on Graph-Based Anomaly Detection: Approaches for Intrusion Detection MDPI, accessed on June 30, 2026, [https://www.mdpi.com/2076-3417/16/4/1906](https://ww
3417/16/4/1906)

- MolGraphBench: A Benchmark of GNN Architectures for Molecular Regression Tasks arXiv, accessed on June 30, 2026, [https://arxiv.org/html/2602.20573v3](https://arxiv.org/html/2602.20

- Explainable AI for Smart Grid False Data Injection Attack Prediction ResearchGate, accessed on June 30, 2026, [https://www.researchgate.net/publication/404392243
$_{E}xplainable$
$_{A}I$
$_{f}or$
$_{S}mart$
$_{G}rid$
$_{F}alse$
$_{D}ata$
$_{I}njection$
$_{A}ttack$
$_{P}rediction](https://www.researchgate.net/publication/404392243_Explainable_AI_for_Smart_Grid_$

- Which Explainable AI Methods Work? LIME, SHAP, Attention Reality Check, accessed on June 30, 2026, [https://www.dataannotation.tech/blog/explainable-ai-methods](https://www.dataann
ai-methods)

- False Data Injection Attack Detection in Smart Grid Based on Learnable Unified Neighborhood-Based Anomaly Ranking MDPI, accessed on June 30, 2026, https://www.mdpi.com/2079-9292/14/17/3396

- Additional Tennessee Eastman Process Simulation Data for Anomaly Detection Evaluation GitHub, accessed on June 30, 2026, [https://github.com/jonathanwvd/awesome-industrial-datasets/blob/master/markdown/additional
$_{t}ennessee$
$_{e}astman$
$_{p}rocess$
$_{s}imulation$
$_{d}ata$
$_{f}or$
$_{a}nomaly$
$_{d}etection$
$_{e}valuation.md$](*https* : *//github.com/jonathanwvd/awesome−industrial−datasets/blob/master/mark*

- Digital Twin Concept, Method and Technical Framework for Smart Meters Semantic Scholar, accessed on June 30, 2026, [https://pdfs.semanticscholar.org/bb2e/f807326ba98e2b2c684fc8ff64d

- Cora Hartmann, accessed on June 30, 2026, [https://ad-publications.cs.uni-freiburg.de/theses/Master
$_{C}ora$
$_{H}artmann$
$_{2}025.pdf$](*https* : *//ad−publications.cs.uni−freiburg.de/theses/Master_Cora_Hartmann_*2025.*pdf*)

- Experimental Validation and Deployment of Observability Applications for Monitoring of Low-Voltage Distribution Grids ResearchGate, accessed on June 30, 2026, [https://www.researchgate.net/
$_{E}xperimental$
$_{V}alidation$
$_{a}nd$
$_{D}eployment$
$_{o}f$
$_{O}bservability$
$_{A}pplications$
$_{f}or$
$_{M}onitoring$
$_{o}f$
$_{L}ow - Voltage$
$_{D}istribution$
$_{G}rids$](*https* : *//www.researchgate.net/publication/*354527967*_Experimental_Validation_and_Deployn*
Voltage_Distribution_Grids)

- Residual-aware health prediction of power transformers via spatiotemporal graph neural networks PMC, accessed on June 30, 2026, [https://pmc.ncbi.nlm.nih.gov/articles/PMC12599914/](https

- A Real-Time Implementation and Validation of Federated Learning for Grid Services, accessed on June 30, 2026, [https://www.researchgate.net/publication/398023911
$_{A}$
$_{R}eal - Time$
$_{I}mplementation$
$_{a}nd$
$_{V}alidation$
$_{o}f$
$_{F}ederated$
$_{L}earning$

$_{f}or$
$_{G}rid$
$_{S}ervices$](*https* : *//www.researchgate.net/publication/*398023911*_A_Real−Time_Implementation_and*

- Review of Design Elements within Power Infrastructure Cyber–Physical Test Beds as Threat Analysis Environments PSE Community.org, accessed on June 30, 2026, [https://psecommunity.or content/plugins/wpor/includes/file/2304/LAPSE-2023.30069-1v1.pdf](https://psecommunity.org/wp-content/plugins/wpor/includes/file/2304/LAPSE-2023.30069-1v1.pdf)
- Real-Time State Estimation in Smart Grids over 5G Networks: Experimental Validation Using Raspberry Pis and Typhoon HIL arXiv, accessed on June 30, 2026, [https://arxiv.org/html/2606.27
- Cybersecurity Assessment of Smart Grid Exposure Using a Machine Learning Based Approach arXiv, accessed on June 30, 2026, [https://arxiv.org/html/2501.14175v1](https://arxiv.org/html/

[image1]: ¡

[image2]: ¡

[image3]: ¡

[image4]: ¡

[image5]: ¡

[image6]: ¡

[image7]: ¡

[image8]: ¡

[image9]: ¡

[image10]: ¡

[image11]: ¡

[image12]: ¡

[image13]: ¡

[image14]: ¡

[image15]: ¡

[image16]: ¡

[image17]: ¡

[image18]: ¡

[image19]: ¡

[image20]: ¡

[image21]: ¡

[image22]: ¡

[image23]: ¡

[image24]: ¡

[image25]: ¡

[image26]: ¡

[image27]: ¡

[image28]: ¡

[image29]: ¡

[image30]: ¡

[image31]: ¡

[image32]: ¡

[image33]: ¡

[image34]: ¡

[image35]: ¡

[image36]: ¡

[image37]: ¡

[image38]: ¡

[image39]: ¡

[image40]: ¡

[image41]: ¡

[image42]: ¡

[image43]: ¡

[image44]: ¡